UNIVERSITY OF CALGARY

Phys 369: Acoustics, Optics and Radiation

Lecture 7: Sep 17, 2026

Last time:

- Transverse vs longitudinal waves
- Travelling waves and the wave equation
- Total phase for sinusoidal waves
- Superposition of waves

Today's class:

- Complex representation for wave addition (interference)
- Waves on a stretched string
- Using dimensional analysis

COMPLEX REPRESENTATION OF A WAVE

Example

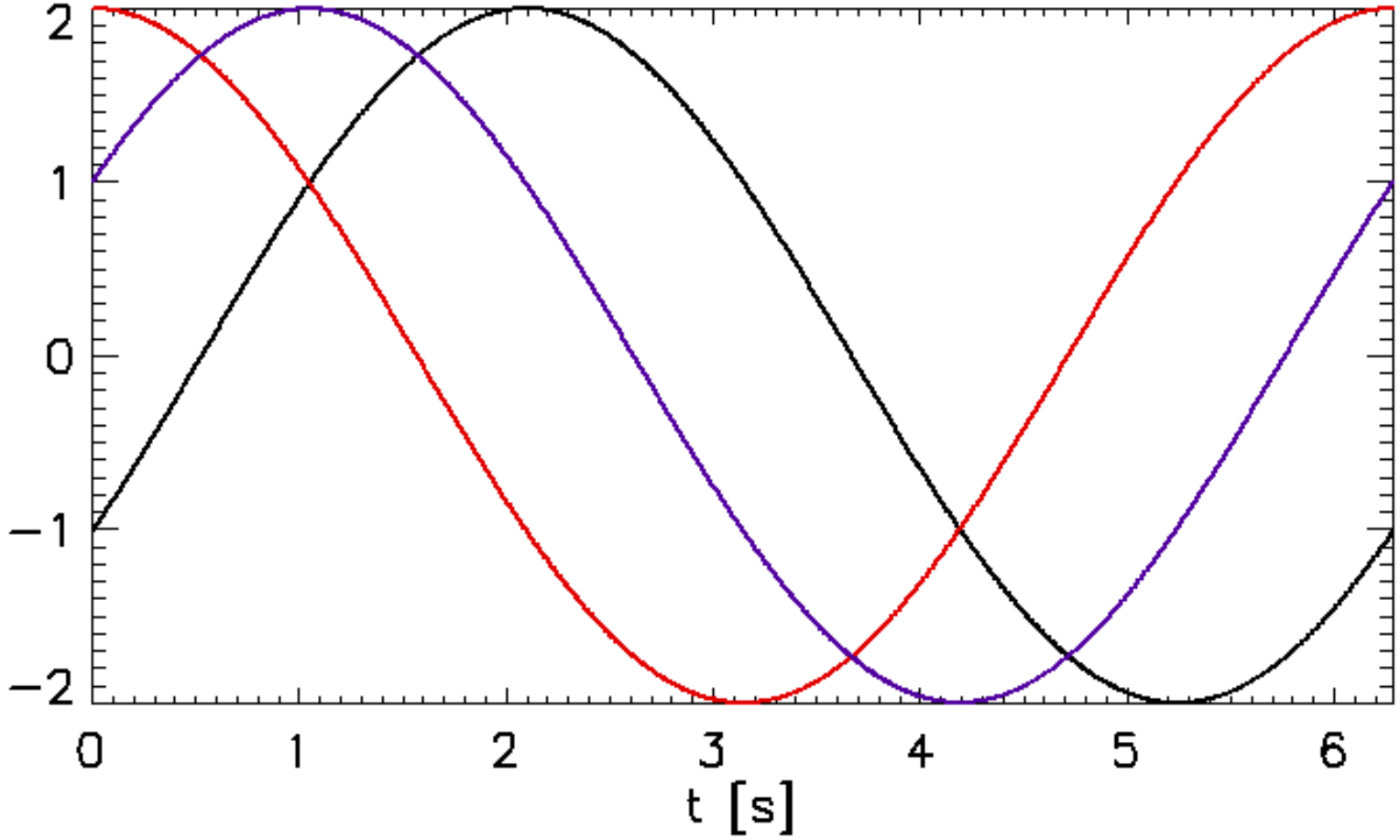
Two waves:

$$y_1(x,t) = (2.0\ \text{m})\sin(kx - \omega t + 7\pi/6)$$
$$y_2(x,t) = (2.0\ \text{m})\cos(kx - \omega t)$$

travel together down a string with ω=1 rad/s.

Find the amplitude and phase of the resulting superposition

$$y_3(x,t) = y_1 + y_2 = A\cos(kx - \omega t + \phi_0)$$



y_1
y_2
$y_3 = y_1 + y_2$

Summary

more info

- The medium does not move at the same speed as the wave. The medium velocity and acceleration are:

$$y = A\cos(kx - \omega t + \phi_0)$$

$$v_y = A\omega\sin(kx - \omega t + \phi_0)$$

$$a_y = -A\omega^2\cos(kx - \omega t + \phi_0)$$

- Wave equation: $\frac{\partial^2 y(x,t)}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y(x,t)}{\partial t^2}$ with $v = \omega/k$
- Waves may be linearly superposed
- Waves can be written in complex notation as $\tilde{y}(x,t) = \tilde{A}e^{j(kx-\omega t)}$

WAVES ON A STRETCHED STRING

EXAMPLE

A 6 m long rope has a mass of 1.0 kg. It is held taut with 12 N of force. How long does it take for a wave to travel from one end to the other?